Ethan Stace

ethanstace@yahoo.com | estace107.github.io

Education

University of Florida August 2023 - May 2027
BS in Physics, Mathematics, Minor in Electrical Engineering (GPA: 3.93/4.00) *Gainesville, FL*

- **Relevant Coursework:** Solid State Physics, Solid State Devices, Statistical Mechanics, Quantum Mechanics I & II, Circuits, Photonics, MEMS Transducers, Electromagnetism I & II, Spintronics, Machine Learning for Physics

Experience

Undergraduate Researcher in Electrical Engineering November 2025 - Present
University of Florida *Gainesville, FL*

- Advisor: Prof. Philip Feng (Electrical & Computer Engineering)
- Developing 4-node Ising machine using NEMS resonators to solve Max-Cut problems
- Characterized and tested NEMS resonators using probe station, lock-in amplifier, and oscilloscope
- Designing PCB to miniaturize Ising machine on KiCad software
- Fabricating thin-film resonators using cleanroom techniques for low-temperature measurements in Microkelvin Laboratory

Penn State 2DCC/Nanofab REU Intern May 2026 - July 2026
The Pennsylvania State University *State College, PA*

- Advisors: Prof. Saptarshi Das, Prof. Suzanne Mohney (Electrical & Computer Engineering)
- Assisted in fabrication and characterization of 2D field-effect transistors using cleanroom and lab equipment. Shadowed electron-beam lithography, electron-beam deposition, atomic layer deposition, and reactive ion plasma etching
- Performed rapid thermal annealment and characterization of ohmic contacts to semiconductor thin films
- Analyzed XRD data of ohmic contacts to AlN using Jade software

UT-Dallas Physics REU Intern May 2025 - August 2025
The University of Texas at Dallas *Dallas, TX*

- Advisor: Prof. Aaron Smith (Astrophysics)
- Developed Atom module in Cosmic Lyman-α Transfer Code, a new module for simulating multi-lined, coupled resonant scattering in astrophysical media
- Developed novel analytical solution to radiative transfer equation to model multi-lined transfer networks
- Used Atom module to study Lyman-α and Deuterium-injected systems

Undergraduate Researcher in Plasma Astrophysics October 2024 - May 2025
University of Florida *Gainesville, FL*

- Advisor: Prof. Siyao Xu (Physics)
- Analyzed data from HPC simulations to study physics of weakly-collisional positron-electron plasma
- Developed novel algorithm to measure mean free path by tracking sharp changes in trajectories, and how the mean free path changed at different length and energy regimes
- Showed that mean free path distribution is non-Maxwellian, matching expected behavior from anisotropy of weakly-collisional plasmas.

Extracurricular Activities

Society of Physics Students | *Outreach Chair, (3x)* Fall 2024 - May 2027

- Lead the outreach efforts of the Society of Physics Students, both within the department and with outside organizations
- Created the outreach committee to brainstorm ideas and organize events
- Lead the Mentor-Mentee program, responsible for Mentor-Mentee speed-dating, setting up social events, and ensuring healthy communication between matches

SCEECS Summer School on Extreme Electrodynamics and Plasma Physics | *Student* June 2025

- Attended intensive two weeks of graduate-level lectures on plasma astrophysics at The University of Texas at San Antonio taught by renowned scientists from academic institutions across the world

SIMIODE Challenge Using Differential Equations Modeling | *Competitor* November 2023

- Joined a three-person team to compete in an international math-modeling competition
- Created infectious disease model using Lotka-Volterra equations to model the spread of violence through human populations, then analyzed the behavior of solutions using numerical methods
- Scored the highest raw score (4.857/5.000) out of all 124 teams, earned *Outstanding Team Award*

Publications, Posters, and Presentations

Publications

- Caplice, T., Kaiser, T., **Stace, E.**, Shaw, S., Casella, C., Feng, P.X.-L. "A Nanoelectromechanical Oscillator Ising Machine for Solving Max-Cut Problems." Nature Microsystems and Nanoengineering. (Submitted)
- **Stace, E.**, Smith, A., Lorinc, K., Nebrin, O. "Multi-Resonant-Line Radiative Transfer: Lyman-α Fine Structure and Deuterium Coupling." Publications of the Astronomical Society of Australia. https://arxiv.org/abs/2511.20580. (Accepted 1/22/26)

Conference Papers

- Caplice, T., Kaisar, T., **Stace, E.**, Roukes, M., Casella, C., Shaw, S., Feng, X. L. "Experimental Demonstration of Nanomechanical Oscillator Ising Machine for Solving Optimization Problems." Hilton Head Workshop 2026.

Presentations

- Randazzo, A., Prince, N., **Stace, E.**, Burrowes, A. "Atomistic Spin Simulations with Sunny.jl." Introduction to Solid State Physics, 2026
- **Stace, E.**, Smith, A., Lorinc, K., Nebrin, O. "Multi-Resonant-Line Radiative Transfer: Lyman-α Fine Structure and Deuterium Coupling." 2026 APS Global Physics Summit.
- **Stace, E.** "A Short Review of the Feynman Path Integral." Introduction to Particle Physics, 2025
- **Stace, E.** "Predicting Wheat Diseases using CNNs." Machine Learning for Physics, 2024

Posters

- **Stace, E.**, Alam, M.D.S., Nam, T., Das, S. "Threshold Engineering for 2D Field-Effect Transistors." 2026 Penn State Research Experience for Undergraduates (REU) Summer Symposium. July 30, 2026
- **Stace, E.**, Smith, A., Lorinc, K. "Leveling Up: Resonant Scattering within Multi-Level Atoms." 2025 Summer Platform for Undergraduate Research (SPUR) Symposium. July 31, 2025
- **Stace, E.**, Xu, S. "Numerical Measurement of Mean Free Path of High-Energy Weakly-Collisional Plasmas." 2025 Annual Physics Undergraduate Research Poster Session. April 25, 2025

Awards

- *Hilton Head Workshop 2026 Award for Conference Paper*: For top 4 papers at the conference
- *CLAS University Scholars Program*: Funded by University of Florida for undergraduate research during 2025-2026 academic year
- *DAP Travel Grant*: Travel reimbursement towards 2026 APS Global Physics Summit
- *Garrett Endowment for Physics Education* (4x)
- *2025 Erik Jonsson School of Engineering and Computer Science Award*: for exceptional poster at SPUR Symposium
- *2023 SIMIODE Challenge Using Differential Equations Modeling (SCUDEM) Outstanding Team Award*
- *Florida Bright Futures Scholarship*: 100% tuition coverage for four years

Technical Skills

Programming Languages: Python, C++, Verilog, Bash, HTML, CSS
Libraries: NumPy, Matplotlib, SciPy, H5Py, Pandas, Tensorflow, Sympy, Sunny.jl
Laboratory Techniques/Equipment: Network Analyzer, Wire Bonding, 4/2 -Point Probe Station, Mechanical Exfoliation, Scanning Electron Microscopy, Raman Spectroscopy, X-Ray Diffraction, Chemical Cleaning
Cleanroom Techniques/Equipment: Spin Benches, Liftoff Bath, Rapid Thermal Annealing, Photolithography
Other Technologies: KiCad, MATLAB, Vivado, Vitis, Quartus, Linux, Git, Github, SLURM, Jade, LaTeX